

AIONOS · ASSIGNMENT 3

AeroDesk

A customer facing airline disruption resolution agent that decides by policy not by guesswork.

Live demo · acgmopwqqrfw7kuguizurk.streamlit.app

Streamlit prototype · Python

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Core Business Requirements

1

Understand intent

Free-text customer messages — cancellation, delay, refund, rebooking, hotel, upgrade.

2

Use only supplied data

Customers, PNRs, flights and policy rules from the data pack. Nothing invented.

3

Decide by policy

Every reply traces back to a named rule — never an improvised compensation amount.

4

Handle frustration

Detects angry / anxious phrasing and leads with an apology before the facts.

5

Execute allowed actions

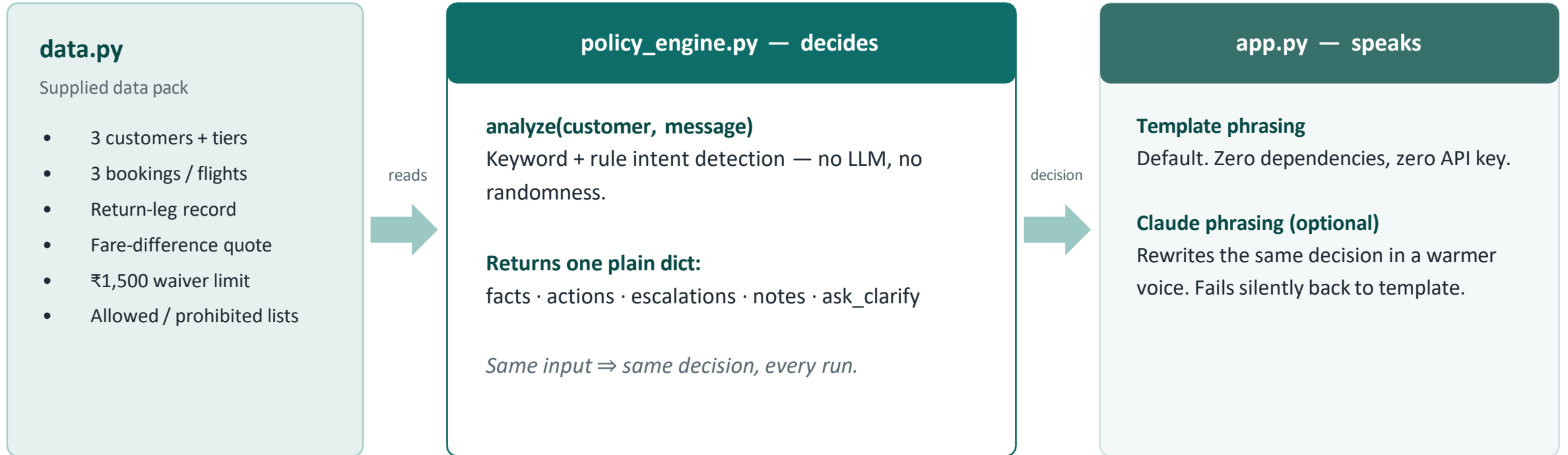
Refund, rebooking, vouchers, lounge, delayed-hours hotel — logged as they happen.

6

Escalate the rest

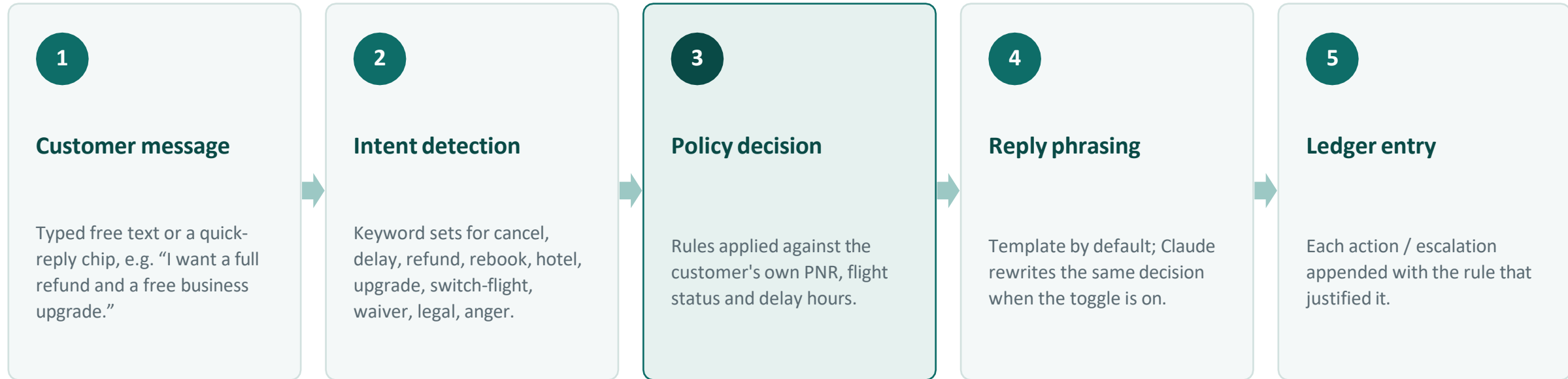
Anything beyond agent authority goes to a human, with the reason recorded.

Two Layer Architecture and Responsibility Model



The LLM never sees the data pack or the rules only the decision that was already made. It cannot approve an escalation, invent an amount, or change an entitlement.

Customer Request to Resolution





Data Scope and Key Assumptions

Inputs

- Customer profile — tier, PNR, email, phone, travel history
- Booking record — flight, route, date, status, delay hours, new departure
- Return leg where the pack lists one (display only)
- Fare-difference quote for a voluntary switch
- The customer's free-text message

Sources

- The assignment data pack only — copied into data.py unmodified
- Policy rules: cancellation rebooking, delay compensation tiers, hotel threshold, ₹1,500 waiver limit, refund processing
- No external API, database or web lookup anywhere in the app
- No customer data leaves the session

Assumptions made

- Meal voucher valued at ₹500, stated consistently rather than left vague
- Refunds go to the original payment method within 7 business days
- “Full night” hotel ≠ delayed-hours hotel, so it needs sign-off
- Gold / Platinum get priority rebooking, not extra compensation
- A quoted fare difference is flagged, but only escalated if a waiver is actually asked for



Policy Driven Decision Matrix

SCENARIO	WHAT THE AGENT DOES	OUTCOME
Cancellation	Free rebooking within 24h or full refund — customer's choice	Allowed
Delay under 3h	Meal voucher (₹500)	Allowed
Delay 3–5h	Meal voucher + lounge access	Allowed
Delay over 5h	Adds hotel for the delayed-hours portion only	Allowed
Fare difference ≤ ₹1,500	Agent may waive	Allowed
Fare difference > ₹1,500	Supervisor approval required	Escalate
Free upgrade “for the trouble”	Compensation beyond policy	Escalate
Full night's hotel stay	Beyond the delayed-hours entitlement	Escalate
Legal action / formal complaint	Handled by a human immediately	Escalate



AI Integration & Governance

AI

Claude (Anthropic API)

In the product

Optional phrasing layer in app.py. Receives only the decision dict, with an explicit instruction not to add facts or approve anything. Model: claude-sonnet-4-6, max 300 tokens, wrapped in try/except with a template fallback.

AI

Claude (chat)

While building

Used as a pair-programmer: drafting the Streamlit layout and CSS, structuring the intent keyword sets, and reviewing edge cases such as the fare-difference flag vs. escalation distinction.

AI

GitHub Copilot in VS Code

While building

Line-level autocompletion across app.py and policy_engine.py, and quick scaffolding of the notebook sanity checks.

Kept out of AI's hands: intent-to-policy mapping, compensation amounts, escalation triggers and the ledger. Pull the API key and the agent still behaves identically.



Representative Customer Scenarios

Priya Nair

GOLD · SK4821X

SK-204 Delhi → Goa cancelled

AGENT ACTS

- Refund or free rebooking within 24h offered
- Priority rebooking noted for Gold tier

AGENT ESCALATES

- Free business-class upgrade “for the trouble”

Arvind Kulkarni

SILVER · TR1190B

SK-118 Mumbai → Bengaluru, 4h delay

AGENT ACTS

- Meal voucher + lounge access issued

AGENT ESCALATES

- Hotel request — delay is under 5h
- “I might file a complaint” — immediate handoff

Meher Kaur

PLATINUM · WL7742

SK-305 Delhi → Hyderabad, 6h delay

AGENT ACTS

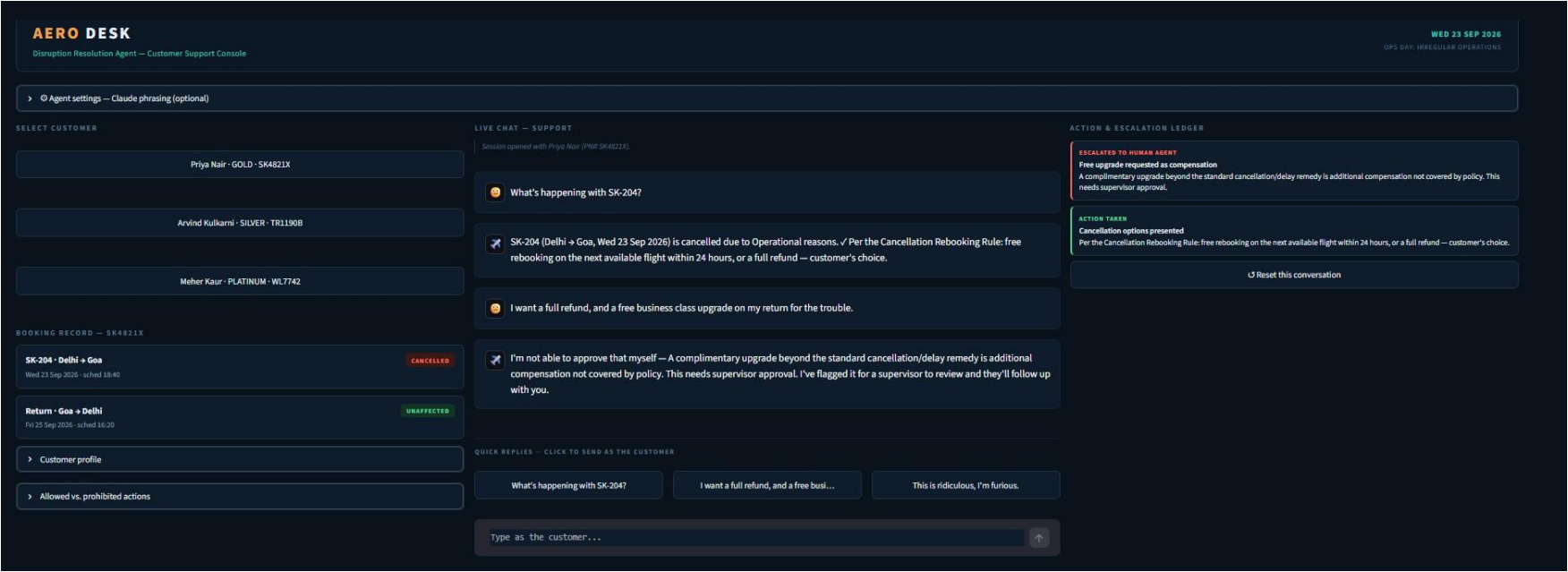
- Voucher, lounge and delayed-hours hotel
- ₹2,000 fare difference quoted to switch

AGENT ESCALATES

- Full night's stay requested
- Waiver of the ₹2,000 difference



Unified Resolution Console



Left context

Customer switcher, booking record with live status badges, profile and the allowed vs. prohibited list.

Centre chat

Free text or quick-reply chips; the agent answers in the customer's own language.

Right ledger

Every action taken and every escalation, each with the policy that justified it.

Live at acqmopwqgrfw7kuqujzurk.streamlit.app ·

Shipped, and where it goes next

5

Python / notebook files

9

policy rules encoded

0

invented compensations

Run it: `streamlit run app.py`

Live demo · acgmopwqgrfw7kugujzurk.streamlit.app

Stack: Python · Streamlit · anthropic SDK (optional) · deployed on Streamlit Community Cloud from GitHub.

Next steps

- LLM-assisted intent parsing behind the same rule gate
- Persist the ledger to a real case-management system
- Live PNR and inventory lookup in place of the static pack
- Supervisor queue with approve / decline

Thank you